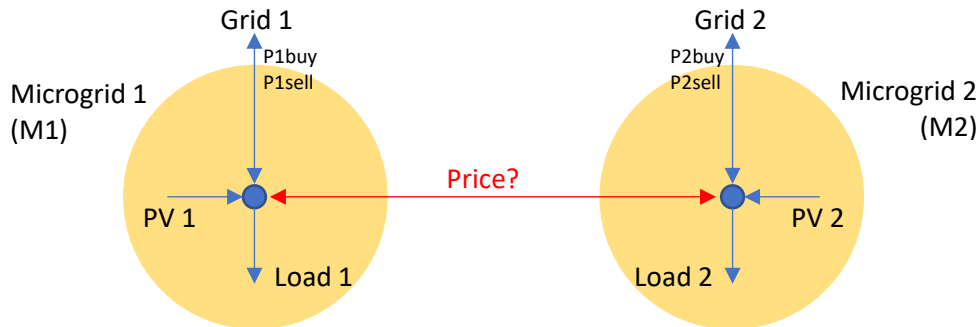


A microgrid can be simplified as a consumer, a generator, and connection to the external electric grid. You can build your own model of microgrid with the connection to the grid modelled just as a price to buy and a price to sell energy.

Then, if there are two microgrids that can exchange power between them, which is the price that they will put on the energy exchange?



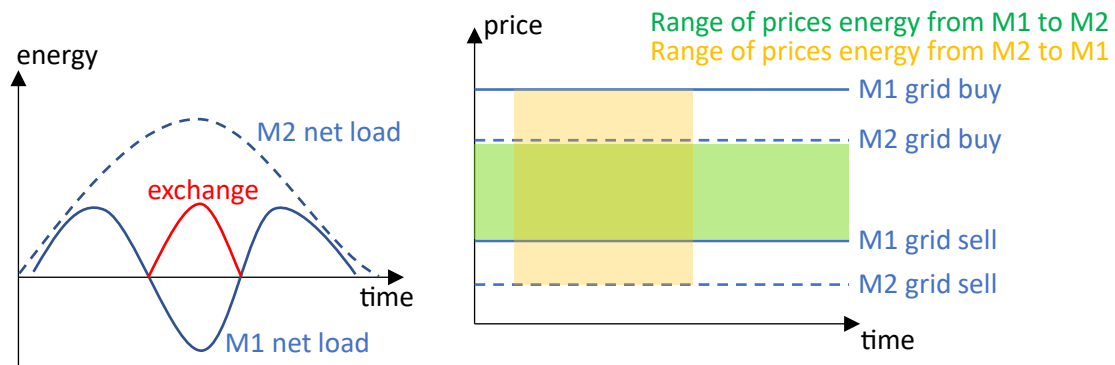
Assumptions:

- The profiles of microgrid 1 and microgrid 2 should be different
- Prices of buy and sell from the grid can also be different

We can analyse different scenarios:

- According to the net load of microgrids (load-PV):
 - Microgrid 1 $\gg$ Microgrid 2
 - Microgrid 1 $\approx$ Microgrid 2
- According to the PV:
 - If PV installed is known
 - If PV should be sized in the optimization
- According to the battery storage:
 - No BESS
 - BESS capacity is known
 - BESS capacity should be sized in the optimization
- According to the number of microgrids that can be interconnected
 - 2 microgrids
 - 3 microgrids
 - 4 microgrids
- According to the optimization approach:
 - The whole system is optimized together, later the power exchange is calculated, and the prices (or range of prices) are calculated
 - Each microgrid is optimized by separate, later the power exchange between microgrids can be calculated or optimized.
 - Other approaches found in literature? Maybe something about game theory?

For example, given two microgrids with the following net load profiles and prices with the grid, the range of prices of the energy exchange from M1 to M2 is shown in green:



Work plan:

1. Search on the literature about interconnecting energy communities or microgrids, and how is defined the price for the energy exchange
2. Build your own microgrid model. And a system with 2 microgrids.
3. Optimize the whole system together and see what happens. This is like putting on the optimization a 0 price on the exchange
4. ...

The objective is to learn what is done on the literature and analyse as many scenarios as possible in the remaining time.